

Practical 2

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Q. 1. Write algorithm for the following:

a. Write a program to generate the Fibonacci series.

Code:

#2A: Program to generate Fibonacci Series up to n terms

```
m=int(input("Enter the number= "))
```

```
#First 2 Terms
```

```
a=0
```

```
b=1
```

```
count=0
```

```
print("Fibonacci Series:")
```

```
while count < m:
```

```
    print(a, end=" ")
```

```
    a, b=b, a + b
```

```
    count += 1
```

Output:

```
===== RESTART: C:/Users/DELL/OneDrive/Desktop/Mel
Enter the number= 7
Fibonacci Series:
0 1 1 2 3 5 8
```

b. Write a program to accept a number from the user display sum of its digits.

Code:

#2B Write a Python program to accept a number and displays the sum of its digits.

```
sum=0
m=int(input("Enter the number= "))

while (m>0):
    digit= m%10
    sum+=digit
    m//=10 # // means to truncate
print("Sum of digits of the given number= ", sum)
```

Output:

```
> |
  | ===== RESTART: C:/Users/DELL/OneDrive/
  | Enter the number= 29
  | Sum of digits of the given number= 11
  |
> |
```

Q. 3. Write a python program to display astrological sign for given date of birth. (Take input: month and day. Check which range the date falls into using if-elif statements. Return or print the corresponding zodiac sign.)

Code:

```
day = int(input("Enter birth day: "))
month = int(input("Enter birth month (1-12): "))

if (month == 3 and day >= 21) or (month == 4 and day <= 19):
    print("Your Zodiac sign is Aries")
elif (month == 4 and day >= 20) or (month == 5 and day <= 20):
```

```
    print("Your Zodiac sin is Taurus")
elif (month == 5 and day >= 21) or (month == 6 and day <= 20):
    print("Your Zodiac sin is Gemini")
elif (month == 6 and day >= 21) or (month == 7 and day <= 22):
    print("Your Zodiac sin is Cancer")
elif (month == 7 and day >= 23) or (month == 8 and day <= 22):
    print("Your Zodiac sin is Leo")
elif (month == 8 and day >= 23) or (month == 9 and day <= 22):
    print("Your Zodiac sin is Virgo")
elif (month == 9 and day >= 23) or (month == 10 and day <= 22):
    print("Your Zodiac sin is Libra")
elif (month == 10 and day >= 23) or (month == 11 and day <= 21):
    print("Your Zodiac sin is Scorpio")
elif (month == 11 and day >= 22) or (month == 12 and day <= 21):
    print("Your Zodiac sin is Sagittarius")
elif (month == 12 and day >= 22) or (month == 1 and day <= 19):
    print("Your Zodiac sin is Capricorn")
elif (month == 1 and day >= 20) or (month == 2 and day <= 18):
    print("Your Zodiac sin is Aquarius")
elif (month == 2 and day >= 19) or (month == 3 and day >= 20):
    print("Your Zodiac sin is Pisces")
else:
    print("Invalid date")
```

Output:

```
> |
   | ===== RESTART: C:/Users/DELL/OneDrive/Desktop/Me
   | Enter birth day: 21
   | Enter birth month (1-12): 6
   | Your Zodiac sin is Cancer
   |
> |
```

Q5. Write a python program to calculate the sum of 3 given numbers. If the values are equal then return thrice of their sum.

Code:

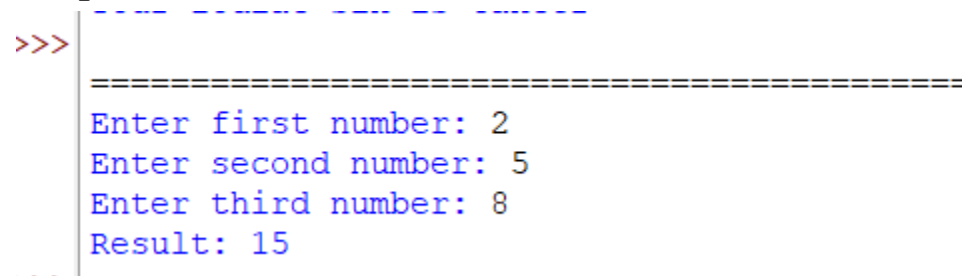
```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
c = int(input("Enter third number: "))
```

```
total = a + b + c
```

```
if a == b == c:  
    print("Result:", total * 3)  
else:  
    print("Result:", total)
```

Output:

```
>>> -----  
=====
```



```
Enter first number: 2  
Enter second number: 5  
Enter third number: 8  
Result: 15
```